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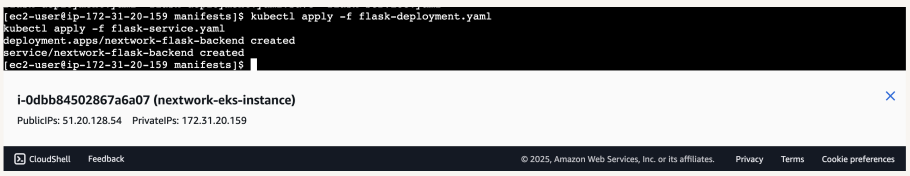
Deploy Backend with Kubernetes

Kehinde Abiuwa



Introducing Today's Project!

In this project, I will set up the backend for deployment, install kubectl, deploy the backend to a Kubernetes cluster, and track the rollout in EKS, because I want a running, observable backend service on Kubernetes that I can monitor and showcase end-to-end DevOps skills.

Tools and concepts

I used Kubernetes (Amazon EKS), Amazon ECR, kubectl, eksctl, and EC2 to create the cluster, push my backend image, map IAM access, and apply Deployment/Service YAML to run and expose the app. Key concepts include using manifests to declare desired state (Deployments & Services), label selectors to route traffic to Pods, replicas for high availability, rolling updates/rollbacks, and Service types (ClusterIP/NodePort/LoadBalancer)

Project reflection

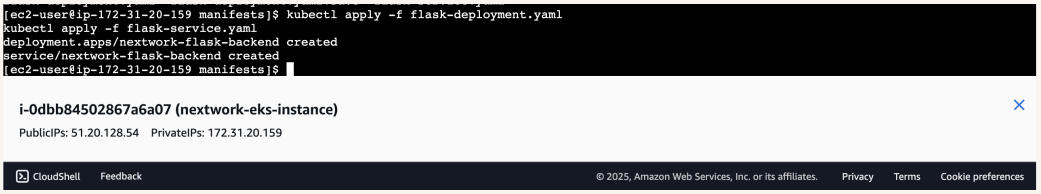
This project took me approximately 40 minutes. The most challenging part was setting up IAM/EKS access so kubectl from EC2 could talk to the cluster and making sure the Service selector/ports matched the Deployment. My favourite part was watching EKS schedule 3 replicas, pull the image from ECR, and start cleanly

Backend Deployment!

To deploy my backend application, I installed kubectl, pointed it at my EKS cluster, and applied my manifests with kubectl apply -f flask-deployment.yaml and kubectl apply -f flask-service.yaml, because that created the Pods from my Deployment and a Service to route traffic to them so the app is reachable

kubectl

kubectl is the Kubernetes CLI that talks directly to the cluster's API server to create, read, update, and delete resources (e.g., Deployment, Service, ConfigMap, Pod). I need this tool to apply my manifests, inspect and debug, and verify the rollout. I can't use eksctl for the job because eksctl is mainly for provisioning and managing EKS clusters. It doesn't replace day-to-day workload operations like applying YAML or interacting with live resources.


[ec2-user@ip-172-31-20-159 manifests]$ kubectl apply -f flask-deployment.yaml
kubectl apply -f flask-service.yaml
deployment.apps/nextwork-flask-backend created
service/nextwork-flask-backend created
[ec2-user@ip-172-31-20-159 manifests]$
i-0dbb84502867a6a07 (nextwork-eks-instance)
PublicIPs: 51.20.128.54 PrivateIPs: 172.31.20.159
CloudShell
Feedback
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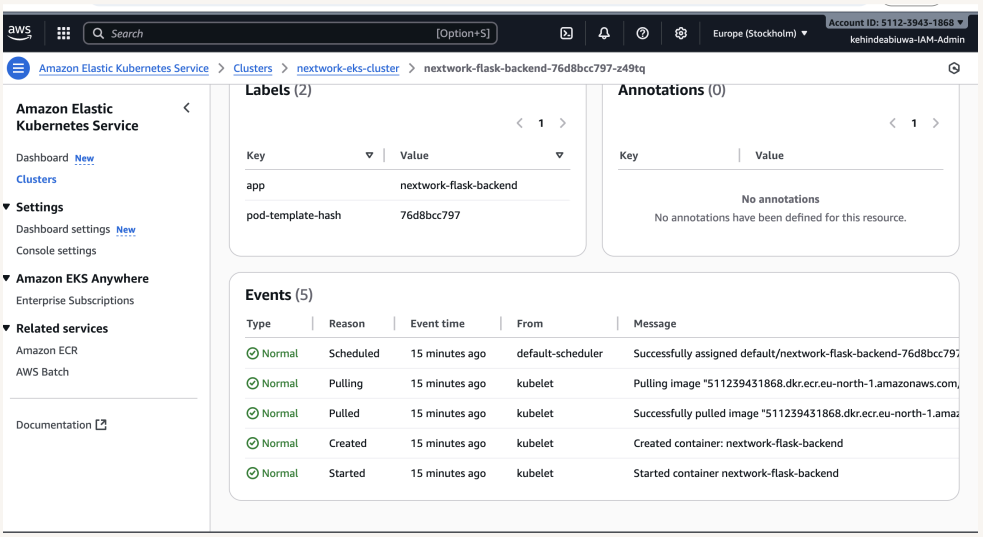
Verifying Deployment

My extension for this project is to use the EKS console to monitor my cluster (nodes, Pods, events) and create an access entry for my EC2 instance. I had to set up IAM access policies because EKS ties AWS IAM to Kubernetes RBAC—without a mapping, kubectl isn't allowed. I set up access by running this in the EC2 Instance Connect tab: eksctl create iamidentitymapping --cluster nextwork-eks-cluster --arn arn:aws:iam::511239431868:user/kehindeabiuwa-IAM-Admin --group system:masters --username admin --region eu-north-1

Once I gained access into my cluster's nodes, I discovered pods running inside each node. Pods are the smallest deployable units in Kubernetes—wrappers around one or more tightly-coupled containers that are scheduled together on the same node and managed as a single unit (via a Deployment/ReplicaSet). Containers in a pod share the same network namespace (one IP and port space, talk over localhost), the same storage volumes, and the same lifecycle/scheduling (they start/stop together).

The EKS console shows you the events for each pod, where I could see it was Scheduled, the node Pulling the ECR image, Pulled successfully, Created the nextwork-flask-backend container, and Started it. This validated that the Deployment worked end-to-end


aws
Search
[Option+S]
Europe (Stockholm)
Account ID: 5112-3943-1868
kehindeabiuwa-IAM-Admin
Amazon Elastic Kubernetes Service
Clusters
nextwork-eks-cluster
nextwork-flask-backend-76d8bcc797-z49tq
Amazon Elastic Kubernetes Service
Dashboard New
Clusters
Settings
Dashboard settings New
Console settings
Amazon EKS Anywhere
Enterprise Subscriptions
Related services
Amazon ECR
AWS Batch
Documentation
Labels (2)
Key
Value
app
nextwork-flask-backend
pod-template-hash
76d8bcc797
Annotations (0)
No annotations
No annotations have been defined for this resource.
Events (5)
Type
Reason
Event time
From
Message
Normal
Scheduled
15 minutes ago
default-scheduler
Successfully assigned default/nextwork-flask-backend-76d8bcc797
Normal
Pulling
15 minutes ago
kubelet
Pulling image "511239431868.dkr.ecr.eu-north-1.amazonaws.com/
Normal
Pulled
15 minutes ago
kubelet
Successfully pulled image "511239431868.dkr.ecr.eu-north-1.amaz
Normal
Created
15 minutes ago
kubelet
Created container: nextwork-flask-backend
Normal
Started
15 minutes ago
kubelet
Started container nextwork-flask-backend

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